

A COMPARISON OF INDIA'S SUDARSHAN CHAKRA WITH ISRAEL'S IRON DOME

Feature	Israel - Iron Dome	India - Mission Sudarshan Chakra
Primary role	Intercepts short-range rockets, artillery, and mortars	Multi-layered defence: intercepts drones, aircraft, ballistic/cruise missiles, stealth and hypersonic threats
Range	4km-70km	30km-400km (depending on subsystem: Akash, MR-SAM, S-400, future Kusha/Chakra)
Operational status	In service since 2011	Some systems are operational (Akash, MR-SAM, S-400); multi-layer and AI-enabled projects in development
Missile types	Tamir interceptor missiles	Multiple: Akash, MR-SAM, Barak-8, S-400, and new indigenous missiles proposed
Radar and sensors	AESA radar, fire-control battle management unit	Integrated phased array radars, AI-enabled command and control network
Mobility	Mobile, truck-mounted launchers	Mix of truck, tracked, fixed, and ship/air force-based platforms
Simultaneous targets	Capable of handling swarm attacks and multiple rockets at once	Designed for multiple targets, high saturation threats, and swarm drone defence
Cost per interceptor	Approx. \$40,000-\$100,000	Varies widely; indigenous missiles are expected to have a lower cost, imported systems like S-400 cost higher
Unique features	Rapid response, combat-proven, and high probability of kill (PK)	Multi-layered, broad-spectrum protection, AI decision support, and future-ready integration

(Source: Author's analysis)

and unpredictable disruptions in global supply chains caused by export controls, varying tariffs, and geopolitical constraints.

Strengthening the domestic industrial base through public-private partnerships, indigenous development of critical technologies and components, and long-term procurement commitments is critical to avoid shortages and schedule overruns.

Mission Sudarshan Chakra: The way ahead and the future

Mission Sudarshan Chakra symbolises a strategic transformation towards a coordinated, multi-layered, nationally controlled system-of-systems designed to remain resilient, flexible, and capable of

neutralising threats while protecting vital military and civilian assets. Unlike single-layered systems such as Israel's Iron Dome, which is optimised for short-range rocket defence, Sudarshan Chakra is designed as a comprehensive multi-layered, multi-domain architecture. In contrast to the proposed US Golden Dome, conceptualised as a global multi-layered, multi-domain unified defensive network, Sudarshan Chakra is tailored to India's geographic scale and resource framework.

The system is being researched, developed, manufactured, and deployed indigenously, harnessing the capabilities of startups, research organisations, academia, and the defence industry. The indigenous

capability demonstrated during Operation Sindoor reinforces the commitment to *Atmanirbharta* (self-reliance) in strategic defence systems.

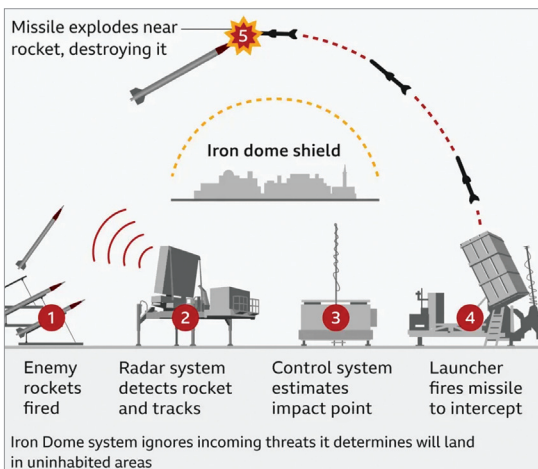
Advanced technologies, including directed-energy systems, hypersonic interceptors, and autonomous counter-drone platforms being developed indigenously, are planned for integration during the second phase. Further improvements in radar sensitivity, sensor

fusion, AI-enabled analytics, cyber resilience, and secure satellite-based communication systems are necessary to reduce response timelines and improve detection and tracking of stealth and hypersonic threats.

Timely availability of supporting facilities such as maintenance hubs, regional training centres, and integrated test ranges is vital. Adaptive logistics frameworks are also important to sustain operations under contested conditions.

Given the scale of integration, technological depth, and automation involved, implementing Sudarshan Chakra within the defined time-frame is a significant undertaking that requires sustained coordination among stakeholders. Availability of specialised manpower, maintenance of technological momentum, expansion of industrial capacity, allocation of sufficient funds (estimated initial expenditure of 16 to 20 billion US\$, with running costs of 3 billion US\$), and institutional coherence must be ensured to prevent schedule overruns.

India has entered the era of AI-enabled warfare. For centuries, decisive advantage came from superior artillery, faster aircraft, or more accurate missiles. Today, the race increasingly centres on which nation can think faster and more accurately with machines. **EFY**



How Israel's Iron Dome defence system works (Source: Rafael Advanced Defense Systems)